

1. The chain rule is a technique for differentiating _____ Matrix No. _____

- (a) composite (b) complex (c) integral (d) None of the option

2. Differentiate $g(x) = \sqrt{x^2 - x + 1}$.

- (a) $g' = \frac{1}{2\sqrt{x^2 - x + 1}}$ (b) $g' = \frac{x-1}{2\sqrt{x^2 - x + 1}}$ (c) $g' = \frac{x^2 - x + 1}{2\sqrt{x^2 - x + 1}}$ (d) None of the option

3. Find $f'(x)$ if $f(x) = \sqrt{x} \cos x$.

- (a) $-\frac{\sqrt{x}}{2}$ (b) $\frac{\sqrt{x}}{2}$ (c) $\frac{\sqrt{x}}{2}$ (d) $g' = \frac{x^2 - x + 1}{2\sqrt{x^2 - x + 1}}$

4. Find the derivative of $\frac{x-1}{x+1}$

- (a) $y' = \frac{x}{(x+1)^2}$ (b) $y' = \frac{1}{(x+1)^2}$ (c) $y' = \frac{-x}{(x+1)^2}$ (d) None of the option

5. The value of $\lim_{x \rightarrow 2} \frac{x+2}{x-4}$ is

- (a) $\frac{1}{4}$ (b) $\frac{1}{4}$ (c) $\frac{1}{2}$ (d) $y' = \frac{x}{(x+1)^2}$

6. Evaluate $\lim_{x \rightarrow \infty} x^2 \sin \frac{1}{x}$

- (a) 0 (b) 1 (c) $\frac{1}{2}$ (d) Undefined

7. Let $f(x) = \begin{cases} x^2 + 1 & \text{if } x \neq 1 \\ 4 & \text{if } x = 1 \end{cases}$

Which of the following statements, I, II and III, are true?

I. $\lim_{x \rightarrow 1} f(x)$ exists

II. $f(1)$ exists

III. f is continuous at $x = 1$.

(a) I and II

(b) I only

(c) II only

(d) None of the Option

8. Suppose $f(x) = \begin{cases} \frac{2x(x-1)}{x^2-3x+2} & \text{for } x \neq 1, 2 \\ f(1) = -3, \\ f(2) = 4. \end{cases}$

Then $f(x)$ is continuous except at

(a) $x = 2$

(b) $x = 1$

(c) $x = 1$ or $x = 2$

(d) $x = 0, 1$, or $x = 2$

9. If h is the inverse of the function of f and if $f(x) = \frac{1}{x}$, then $h(3) =$

(a) $-\frac{1}{3}$

(b) -3

(c) $\frac{1}{3}$

(d) 3

10. If $f(x) = x^{\sin x}$ for $x > 0$, then $f'(x) =$

(a) $x^{\sin x} \left[\frac{\sin x}{x} + (\cos x)(\ln x) \right]$

(b) $x^{\sin x} (\cos x)(\ln x)$

(c) $x \cos x + \sin x$

(d) $(\sin x)^{\sin x - 1}$

11. If $\frac{dy}{dx} = (2x+1)^2$, find y .

- ☒ (a) $\frac{4x^3}{3} + 2x^2 + x + C$ (b) $\frac{4x^3}{3} - 2x^2 - x + C$ (c) $\frac{4x^3}{3} - 2x^2 + x + C$ (d) $\frac{4x^3}{3} + 2x^2 - x + C$

12. Find an equation of the tangent line to the graph of $f(x) = 2x + \ln x$ at $(1, 2)$

- (a) $y = 2 + \frac{1}{x}$ (b) $y = x + 1$ (c) None of the option ☒ (d) $y = 3x - 1$

13. The domain of the exponential function is

- ☒ (a) $(-\infty, \infty)$ (b) $(0, \infty)$ (c) $(0, 1)$ (d) Undefined

14. How do we determine where the derivative of a function $f(x)$ is increasing?

- (a) If $f''(x) < 0$ (b) If $f''(x) = 0$ (c) If $f'(x) = 0$ ☒ (d) If $f''(x) > 0$

15. Inflection points are points where the graph changes _____

- ☒ (a) concavity (b) derivative (c) velocity (d) magnitude

16. Find the intervals on which the function is concave up: $g(x) = x^3 - 3x^2 - 9x + 1$

- ☒ (a) $(1, \infty)$ (b) $(1, 10)$ (c) None of the option (d) $(-\infty, 1)$

17. Find the coordinates of any inflection points of the function $h(x) = x^{\frac{1}{3}}$.

- ☒ (a) $(0, 0)$ (b) $(0, \infty)$ (c) $(-\infty, 0)$ (d) $(0, 1)$

18. Find y' if $f(x) = \frac{4x+3}{2x-1}, x \neq \frac{1}{2}$

- (a) $\frac{10}{(2x-1)^2}$ (b) $\frac{5}{(2x-1)^2}$ (c) $\frac{-5}{(2x-1)^2}$ ☒ (d) $\frac{-10}{(2x-1)^2}$

19. The chain rule is a technique for differentiating _____ functions.

- ☒ (a) composite (b) complex (c) integral (d) None of the option

20. Differentiate $g(x) = \sqrt{3x^2 - x + 1}$.

- (a) $g' = \frac{1}{2\sqrt{3x^2-x+1}}$ (b) $g' = \frac{3x^2-x+1}{2\sqrt{6x-1}}$ (c) $g' = \frac{6x}{2\sqrt{3x^2-x+1}}$ ☒ (d) $g' = \frac{6x-1}{2\sqrt{3x^2-x+1}}$

21. Find $f'(\pi)$ if $f(x) = \sqrt{x} \cos x$.

- (a) $\frac{\sqrt{\pi}}{\pi}$ (b) $-\frac{\sqrt{\pi}}{\pi}$ (c) $\frac{\sqrt{\pi}}{2}$ (d) None of the option

22. Find the derivative of $\frac{t^2-1}{t^2+1}$

- (a) $y' = \frac{5t}{(t^2-1)^2}$ (b) $y' = \frac{t}{(t^2+1)}$ (c) $y' = \frac{-4t}{(t^2+1)^2}$ ☒ (d) $y' = \frac{4t}{(t^2+1)^2}$

23. The value of $\lim_{x \rightarrow -2} \frac{x+2}{x^2-4}$ is

- ☒ (a) $-\frac{1}{4}$ (b) $\frac{1}{4}$ (c) $\frac{-1}{2}$ (d) Undefined

24. Evaluate $\lim_{x \rightarrow \infty} x^2 \sin \frac{1}{x}$

- (a) 1 (b) ∞
(c) oscillates between -1 and 1 ☒ (d) 0

25. Let $f(x) = \begin{cases} x^2 + 1 & \text{if } x \neq 1, \\ 4 & \text{if } x = 1 \end{cases}$

Which of the following statements, I, II and III, are true?

- I. $\lim_{x \rightarrow 1} f(x)$ exists II. $f(1)$ exists III. f is continuous at $x = 1$.

- ☒ (a) I and II (b) I only (c) II only (d) None of the option

Find the value of $\int_0^1 (4x^3 - 3x + 7) dx$.

(a) 7

(b) 7

(c) 7

Find the area bounded by the curve $y = x^2$, the line $y = 2$, and the line $x = 3$.

(a) $\frac{1}{3}$ square unit

(b) $\frac{1}{3}$ square unit

(c) $\frac{1}{3}$ square unit

If $2x = (2x + 1)^2$, find y .

(a) $y^2 = 2x^2 + x + C$

(b) $y^2 = 2x^2 - x + C$

(c) $y^2 = 2x^2 + x + C$

(d) $y^2 = 2x^2 - x + C$

Inflection points are points where the graph changes _____

(a) concavity

(b) derivative

(c) velocity

(d) magnitude

Find the intervals on which the function is concave up. $f(x) = x^3 - 3x^2 - 3x + 1$

(a) $(-\infty, 1)$

(b) $(1, \infty)$

(c) $(1, 10)$

(d) None of the options

Find the coordinates of any inflection points of the function $h(x) = x^3$.

(a) $(0, 0)$

(b) $(0, \infty)$

(c) $(-\infty, 0)$

(d) $(0, 1)$

Find y' if $f(x) = \frac{2x}{x^2 - 1}$, $x \neq \pm 1$

(a) $\frac{2(1-x^2)}{(x^2-1)^2}$

(b) $\frac{2x}{(x^2-1)^2}$

(c) $\frac{2}{(x^2-1)^2}$

(d) $\frac{2x}{(x^2-1)^2}$

Find an equation of the tangent line to the graph of $f(x) = 2x + \ln x$ at $(1, 2)$

(a) $y = 2 + \frac{1}{x}$

(b) $y = 2x - 1$

(c) $y = x + 1$

(d) None of the options

The domain of an exponential function is

(a) $(0, \infty)$

(b) $(0, 1)$

(c) $(-\infty, \infty)$

(d) Undefined

How do we determine where the derivative of a function $f(x)$ is increasing?

(a) If $f''(x) < 0$

(b) If $f''(x) = 0$

(c) If $f''(x) = 0$

(d) If $f''(x) > 0$

18. The integral of a function whose numerator is the derivative of the denominator, is equal to the
 (a) logarithm of the numerator ☐ (b) logarithm of the denominator
 (c) exponent of its derivative ☒ (d) derivative of its logarithm
19. Find the area bounded by the curve $x = y^2$, y -axis and the lines $y = 0$, $y = 2$.
 (a) $\frac{8}{3}$ square unit ☒ (b) $\frac{8}{3}$ square unit
 (c) $\frac{8}{3}$ square unit (d) $\frac{8}{3}$ square unit
20. The chain rule is a technique for differentiating _____ functions.
 (a) complex ☒ (b) composite
 (c) integral (d) Non of the options
21. Differentiate $g(x) = \sqrt{3x^2 - x + 1}$.
 (a) $g' = \frac{1}{2\sqrt{3x^2 - x + 1}}$ (b) $g' = \frac{3x^2 - x + 1}{2\sqrt{6x - 1}}$ ☒ (c) $g' = \frac{6x - 1}{2\sqrt{3x^2 - x + 1}}$ (d) $g' = \frac{6x}{2\sqrt{3x^2 - x + 1}}$
22. Find $f'(\pi)$ if $f(x) = \sqrt{x} \cos x$.
 (a) $-\frac{\sqrt{\pi}}{\pi}$ (b) $\frac{\sqrt{\pi}}{2}$ (c) $\frac{\sqrt{\pi}}{\pi}$ (d) None of the options
23. Find the derivative of $\frac{t^2 - 1}{t^2 + 1}$
☒ (a) $y' = \frac{5t}{(t^2 - 1)^2}$ (b) $y' = \frac{t}{(t^2 + 1)}$ ☒ (c) $y' = \frac{4t}{(t^2 + 1)^2}$ (d) $y' = \frac{-4t}{(t^2 + 1)^2}$
24. The value of $\lim_{x \rightarrow -2} \frac{x+2}{x^2-4}$ is
 (a) $\frac{1}{4}$ (b) $\frac{-1}{2}$ ☒ (c) $\frac{-1}{4}$ (d) Undefined
25. All are techniques of integration except integration by
 (a) substitution (b) parts ☒ (c) limits (d) partial fractions
26. What is the value of $\int_0^2 (4x^2 - 5x + 7) dx$.
 (a) $\frac{40}{3}$ (b) $\frac{44}{5}$ ☒ (c) $\frac{44}{3}$ (d) $\frac{40}{5}$

11. Find the derivative of $y = (4x+1)^{-1}(1-x)^{-1}$.
- (a) $5(4x+1)(1-x)^2(1-4x)$ (b) $(4x+1)^2(1-x)^2(5-20x)$
 (c) $-24(4x+1)(1-x)^2$ (d) $(4x+1)(1-x)^2(4x+11)$
12. If $\sin(xy) = x$, then $\frac{dy}{dx} =$
- (a) $\frac{\sec(xy)-y}{x}$ (b) $\sec(xy)$ (c) $\frac{\sec(xy)}{x}$ (d) None of these
13. Evaluate $\lim_{x \rightarrow 0^+} x^x$
- (a) 1 (b) $\frac{1}{e}$ (c) 0 (d) e
14. A relative minimum value of the function $y = \frac{e^x}{x}$ is
- (a) 0 (b) $\frac{1}{e}$ (c) e (d) 1
15. The slope of the curve $y^3 - xy^2 = 4$ at the point where $y = 2$ is
- (a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $-\frac{1}{2}$ (d) -2
16. Find the point of discontinuity of the function $f(x) = \frac{x^2-3x+6}{x^2-4x+4}$
- (a) 2 (b) 3 (c) 4 (d) 5
17. The standard result for $\lim_{x \rightarrow 0} \frac{a^x-1}{x-a}$ is
- (a) $\log_e a$ (b) 1 (c) $\ln(x-1)$ (d) $\frac{a^x}{x-a}$
18. Evaluate $\int e^{3x+7} dx$
- (a) $\frac{1}{3}e^{3x+7}$ (b) $\frac{1}{7}e^{5x+7}$ (c) $\frac{1}{5}e^{5x+6}$ (d) $\frac{1}{7}e^{4x+6}$
19. Find the integral of $\frac{2x}{x^2+1}$ with respect to x
- (a) $\log(x^2+1) + C$ (b) $\log(x^2+1)$ (c) $\log x^2 + C$ (d) $\log x^2$
20. Evaluate $\int x^2 \ln x dx$
- (a) $\frac{x^2}{3}(\ln x - \frac{1}{3}) + C$ (b) $\frac{x^2}{3}(\ln x + \frac{1}{3}) + C$ (c) $\frac{x^3}{3}(\ln x + \frac{1}{3}) + C$ (d) $\frac{x^3}{3}(\ln x) + C$
21. Evaluate $\int \frac{5+x}{(1-x)(5+x^2)} dx$
- (a) $\ln(x-1) - \frac{1}{2} \ln(5+x^2) + C$ (b) $\frac{1}{2} \ln(5+x^2) + C$
 (c) $\ln(x-1) + \frac{1}{2} \ln(5+x^2) + C$ (d) $\ln(x-1) - \ln(5+x^2) + C$
22. The reduction formula for $\int x^n e^x dx$ is $I_n =$
- (a) $x^n e^x - nI_{n-1}$ (b) $x^n e^x - nI_n$ (c) $x^n e^x - nI_{n-2}$ (d) $x^n e^x - I_{n-1}$
23. What is the value of $\int x^3 e^x dx$
- (a) $e^x(x^3 - 3x^2 - 6x + 6) + C$ (b) $e^x(x^3 - 3x^2 - 6x - 3) + C$
 (c) $x^3 - 3x^2 - 6x + 6 + C$ (d) $e^x(x^3 - 3x^2 - 5x + 6) + C$
24. The integral of a function whose numerator is the derivative of the denominator, is equal to the
- (a) logarithm of the denominator (b) logarithm of the numerator
 (c) exponent of its derivative (d) derivative of its logarithm
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